



Segmented Incremental Sieve

The Segmented Sieve and Incremental Sieve are two algorithms used for finding prime numbers in a certain range.

The Segmented Sieve algorithm is used to find all prime numbers within a given range $[L, R]$ where L and R are two non-negative integers such that $L \leq R$. The basic idea behind this algorithm is to use the Sieve of Eratosthenes method to find all prime numbers up to the square root of R , and then use these primes to eliminate all composite numbers in the range $[L, R]$.

The Incremental Sieve algorithm, on the other hand, is used to find all prime numbers starting from a given number N . The basic idea behind this algorithm is to use the Sieve of Eratosthenes method to find all prime numbers up to a certain limit, say L , and then use these primes to check if $N + i$ is a prime number for $i = 0, 1, 2, \dots$.

Both of these algorithms are efficient for finding prime numbers in certain ranges. The Segmented Sieve is particularly useful for finding primes in large ranges, while the Incremental Sieve is useful for finding primes starting from a given number.

```

public class Segmented {
    static void SegSieve(int l,int h)
    {
        //step1
        boolean prime[] = new boolean[h + 1];
        //step 2
        for (int p = 2; p * p <= h; p++) {
            int sm=(l/p)*p;
            if (sm < l)
            {
                sm=sm+p;
            }
            for (int i = sm; i <= h; i += p)
                prime[i] = true;
        }
    }
}

```

```

        //step 3 Print all prime numbers
        for (int i = l; i <= h; i++)
        {
            if (prime[i] == false)
                System.out.print(i + " ");
        }
    }

    public static void main(String[]
args)
    {
        SegSieve(10, 30);
    }
}

```



THANK YOU